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Logic Correction

Programming Logic · Week 11

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Learning Outcomes

By the end of this session, you can find and fix logic errors in an algorithm.

Explain

Syntax, runtime, and logic errors

Find

Bugs using trace tables and test cases

Fix

Flawed pseudocode and confirm the result

Types of Errors

Aspect	Syntax Error	Runtime Error	Logic Error
Cause	bad grammar	bad operation	wrong idea
Found	before running	while running	by testing
Output	none	crash	wrong answer
Sample	missing END IF	divide by 0	wrong operator
Fix	easy	medium	hard
Tool	compiler	error message	trace table

```
Expected output: 15
Actual output:   10
Runs fine, but the answer is wrong
```

Debugging Steps

STEP 01

Reproduce

Run it with an input that gives a wrong result.

STEP 02

Trace

Follow every variable line by line.

STEP 03

Fix and retest

Change one thing, then test again.

```
IF score >= 60 THEN
    grade = "C"
ELSE IF score >= 80 THEN
    grade = "A"
END IF
// Fix: test score >= 80 first
```

Bug 1: Wrong Order

Every score of 60 or more matches the first condition, so nobody gets an A. Test the highest range first.

```
// Goal: sum of 1 to 5 (answer 15)
total = 0
FOR i = 1 TO 4 DO
    total = total + i
END FOR
OUTPUT total   // prints 10
// Fix: FOR i = 1 TO 5
```

Bug 2: Off-by-One

The loop stops one round early, so the result is 10 instead of 15. Check the upper limit against the goal.

```
total = 0
FOR row = 1 TO 3 DO
    FOR col = 1 TO 4 DO
        total = total + col
    END FOR
    OUTPUT total
END FOR
// Fix: move total = 0 into the outer loop
```

Bug 3: Missing Reset

Each row total should be 10, but the program prints 10, 20, 30 because total is never reset for a new row.

Debugging Checklist

Check 1

Read the goal

Write the expected output first, then compare.

Check 2

Test cases

Use normal, boundary, and invalid inputs.

Check 3

Boundaries

Try values just below, on, and above each limit.

Check 4

Trace by hand

Follow each variable in a trace table.

Check 5

One change

Fix one thing at a time to know what worked.

Check 6

Retest all

After a fix, rerun every earlier test case.

Find and Fix Problem

This program should print the largest of 5 numbers, but it gives a wrong result.

```
max = 0
FOR i = 1 TO 5 DO
    INPUT x
    IF x > max THEN max = x
END FOR
OUTPUT max
```

Tasks

- Trace it with the inputs -4, -2, -9, -1, -7
- Find the logic error and explain why the output is wrong
- Write the corrected pseudocode
- Create 3 test cases that prove your fix

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